山东大学
SHANDONG UNIVERSITY

宣传海报

第 九 届

# HCP 2026

## 难解问题的理论、算法与应用研讨会

2026年7月31日-8月2日　山东大学青岛校区

## 特邀报告

特邀报告 **戴彧虹**

中国科学院数学与系统科学研究院

### 大规模设施选址问题的精确求解

报告摘要

In this talk, we develop exact solution methods for large-scale facility location problems. First, we present algorithms based on decomposition techniques, including Benders decomposition and submodular decomposition, and apply them to large-scale probabilistic set covering and competitive facility location problems. We introduce customized cutting planes and efficient separation procedures, and demonstrate how these algorithms can be integrated into modern mixed-integer programming (MIP) solvers to obtain exact solutions efficiently for large-scale problems.
Second, we develop tailored preprocessing and cutting plane techniques for covering location problems. These techniques can be incorporated into modern MIP solvers' framework either a priori or dynamically during the search, enabling the efficient solution of large-scale covering location problems.

特邀报告 **詹乃军**

北京大学计算机学院

### On termination of polynomial programs with equality conditions

报告摘要

We investigate the termination problem of a family of multi-path polynomial programs (MPPs) over an effective field K, in which all assignments to program variables are polynomials, and test conditions of loops and conditional statements are polynomial equalities. We show that the set of non-terminating inputs (NTI) of such a program is algorithmically computable, which in turn yields the decidability of its termination on a given input - and that on a semi-algebraic set of inputs when K is R. To the best of our knowledge, the considered family of MPPs is hitherto the largest fragment of nonlinear programs for which termination is decidable. We present an explicit recursive function, essentially of Ackermannian growth, to compute the maximal length of ascending chains of polynomial ideals under a control function, thereby providing a complete answer to the questions raised by Seidenberg. This maximal length facilitates a precise complexity analysis of our algorithms for computing the NTI and deciding termination of MPPs. We further extend our approach to programs with polynomial guarded commands and show how an incomplete procedure for MPPs with inequality guards can be obtained. Finally, we show that our decidability result gives rise to a complete method for computing all polynomial equality invariants (of a fixed degree) of polynomial programs.

特邀报告 **孙晓明**

中国科学院计算技术研究所

### 量子线路优化

报告摘要

量子计算依托量子叠加、纠缠等独特量子力学效应构建新的计算范式，Shor算法、Grover算法等显示其在特定计算任务上具备超越经典计算的加速潜力。现阶段量子硬件普遍存在量子比特规模有限、相干时间短、门噪声显著等瓶颈，制约各类量子算法在含噪声中规模量子设备上的稳定、高效执行。量子线路综合与优化作为量子编译的重要环节，是缓解硬件噪声约束、提升量子程序运行保真度的关键技术。本报告将梳理线路优化领域的一些核心问题，并汇报课题组在线路化简、硬件适配映射、深度压缩等方向的一些进展，以及衍生出的优化问题。

特邀报告 **冯启龙**

中南大学计算机学院

### 面向大规模数据的机器学习算法优化

报告摘要

随着数据规模的快速增长，传统机器学习算法在计算开销、动态更新和数据约束等方面面临巨大挑战。本报告主要介绍大规模数据场景下机器学习算法优化方法，重点讨论聚类与回归问题的高效算法设计与分析。针对聚类问题，介绍近线性时间算法、增量学习算法，面向动态更新的算法，并讨论如何去除诸多算法依赖的离散度因子。面向数据约束挑战，介绍带公平性约束的模型与算法，以及线聚类等结构化聚类问题的优化思路。在数据维度处理方面，从子集选择角度出发，介绍高效数据降维算法的设计。针对回归问题，从coreset构造和局部搜索等角度出发，介绍如何进行稀疏数据表示和高效算法设计。